

MOIA

Smooth operator: A remote-operator shift module for MATSim

Simulating remote assistance of autonomous ride-pooling fleets

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MATSim User Meeting, Sep. 28th 2026, Paris



Recap: MUM 2021



Shifts in Perspective: Modeling Operational Challenges in Non-Autonomous and Electric Ride-Pooling Systems

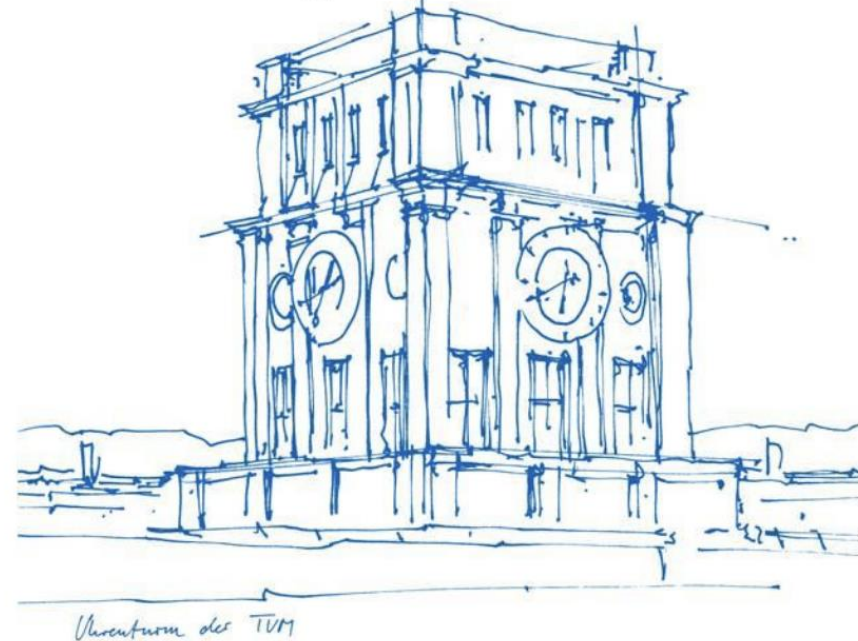
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Assistant Professorship of Modeling Spatial Mobility

MATSim User Meeting, March 22, 2021



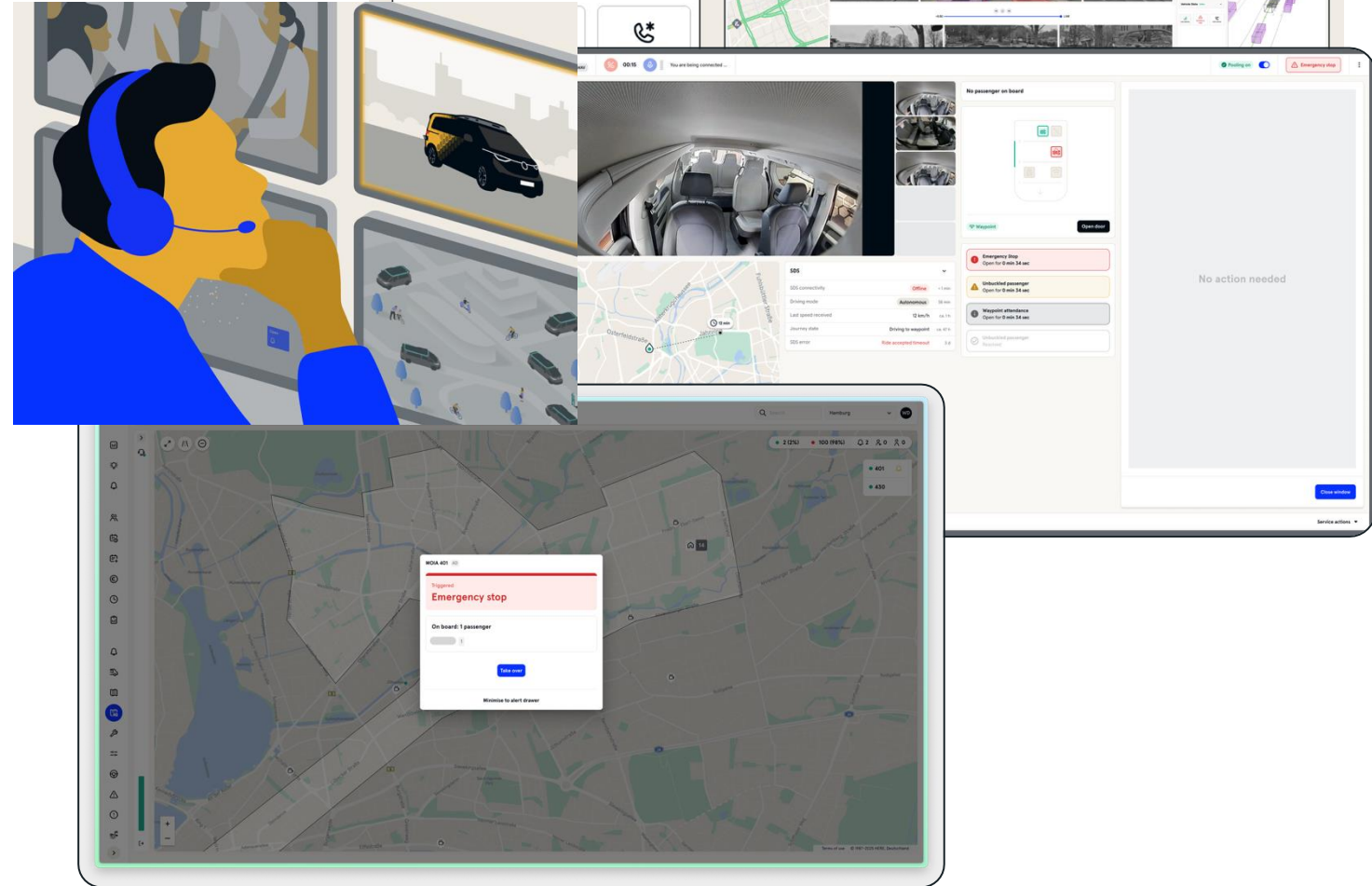
Fast forward to MUM 2026

Multiple vendors have started commercial operations for level 4 autonomous services, many are piloting.

Most if not all of them rely on some form of human supervision.

Just building situation awareness may take a remote operator ~29 s when the vehicle states the cause, ~162 s when it cannot.

Scholtz, J., Antonishek, B., Young, J. (2004). Operator interventions in autonomous off-road driving: effects of terrain. IEEE SMC 2004, 2797–2802. <https://doi.org/10.1109/ICSMC.2004.1400756>
 Mutzenich, C., Durant, S., Helman, S., Dalton, P. (2021). Updating our understanding of situation awareness in relation to remote operators of autonomous vehicles. Cogn. Res. 6, 9



§14 AFGBV in Germany

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Verordnung zur Genehmigung und zum Betrieb von Kraftfahrzeugen mit autonomer Fahrfunktion in festgelegten Betriebsbereichen (Autonome-Fahrzeuge-Genehmigungs- und-Betriebs-Verordnung - AFGBV)

§ 14 Anforderungen an die Technische Aufsicht

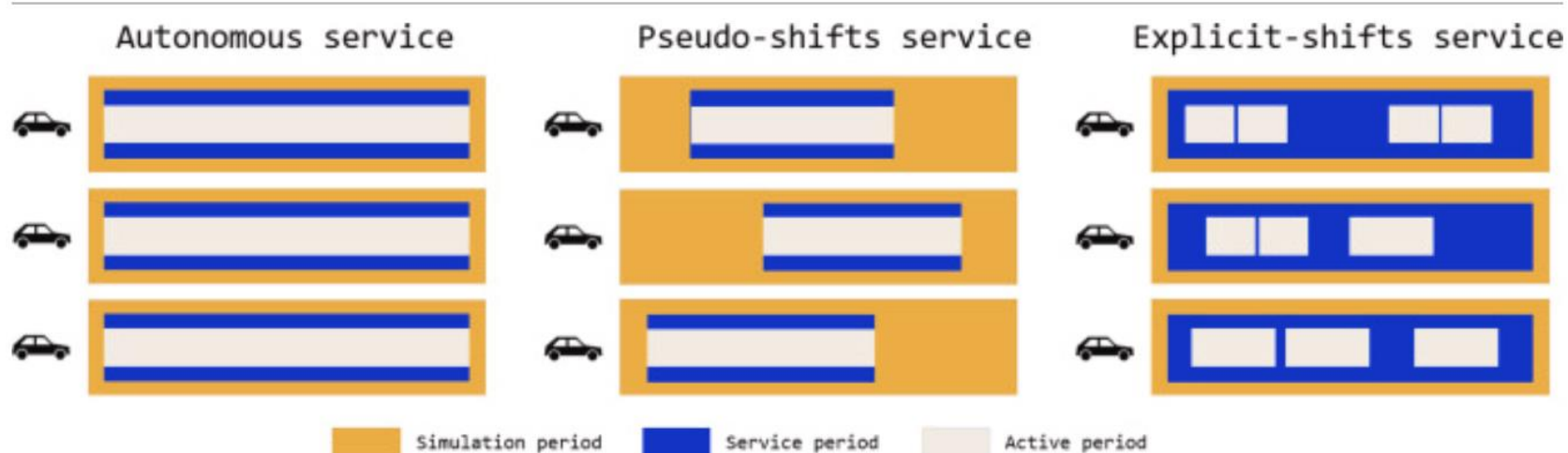
(1) Die als Technische Aufsicht eingesetzte natürliche Person muss für die Wahrnehmung ihrer Aufgaben nach § 1f Absatz 2 des Straßenverkehrsgesetzes geeignet sein. Sie ist geeignet, wenn sie

1. in der Fachrichtung Maschinenbau, Fahrzeugtechnik, Elektrotechnik, Luft- und Raumfahrttechnik oder Luftfahrzeugtechnik über einen Abschluss verfügt als
 - a) Diplom-Ingenieur, Diplom-Ingenieur (FH), Ingenieur (graduiert),
 - b) Bachelor, Master oder
 - c) staatlich geprüfter Techniker,
2. eine entsprechende Schulung in Bezug auf das Kraftfahrzeug mit autonomer Fahrfunktion beim Hersteller dieses Kraftfahrzeugs erfolgreich abgeschlossen hat,
3. einen gültigen Führerschein besitzt, wobei die Klasse der Fahrerlaubnis der des Kraftfahrzeugs mit autonomer Fahrfunktion entsprechen muss, und
4. im Hinblick auf die Wahrnehmung der ihr anvertrauten Aufgaben nach § 1f Absatz 2 des Straßenverkehrsgesetzes zuverlässig ist.

- Level 4 autonomous driving requires „technical supervision“ (in our model: the remote operator) for driving in a permitted ODD
- Among others, staff has to
 1. hold an engineering degree or a state-certified technician qualification
 2. Have a specific training for the operation of the vendor's autonomous system
 3. Hold a valid driver's license
- Remote operators have to ensure that safe continuation after incidents is possible and investigate and document every minimal risk condition, its trigger and whether it was necessary
- Remove vehicle from traffic if safe continuation is not possible

The driver is not removed, it is displaced

- **Robotaxi fleets still depend on people.** Remote operators supervise vehicles and step in when the automated driving system meets a situation outside its operational design domain.
- **One operator, many vehicles (1:N).** Labour is restructured. Reported supervision ratios Waymo $\approx 1:43$, Pony.ai $\approx 1:20$, WeRide $\approx 1:3$, summarised by Abreu Marques (2026); Kaplan et al. (2024).
- **The open planning question:** how many operators does a fleet need, and what does too few do to the service?
- **Simulation so far knows** drivers with shifts of varying detail, or autonomous vehicles free of any operational duty.



Zwick, F., Kuehnel, N., Hörl, S. Shifts in perspective: Operational aspects in (non-) autonomous ride-pooling simulations. Transportation Research Part A: Policy and Practice <https://doi.org/10.1016/j.tra.2022.09.001>

Remote operations are the missing middle

Driver shifts

Zwick, Kuehnel & Hörl (2022)

One driver per vehicle
Shifts, breaks, hub returns
Capacity = drivers on shift

Remote operations

this module

One operator per κ vehicles
Stochastic incidents occupy operators
Capacity = $\kappa \times$ operators on shift

Idealised autonomous

standard AV assumption

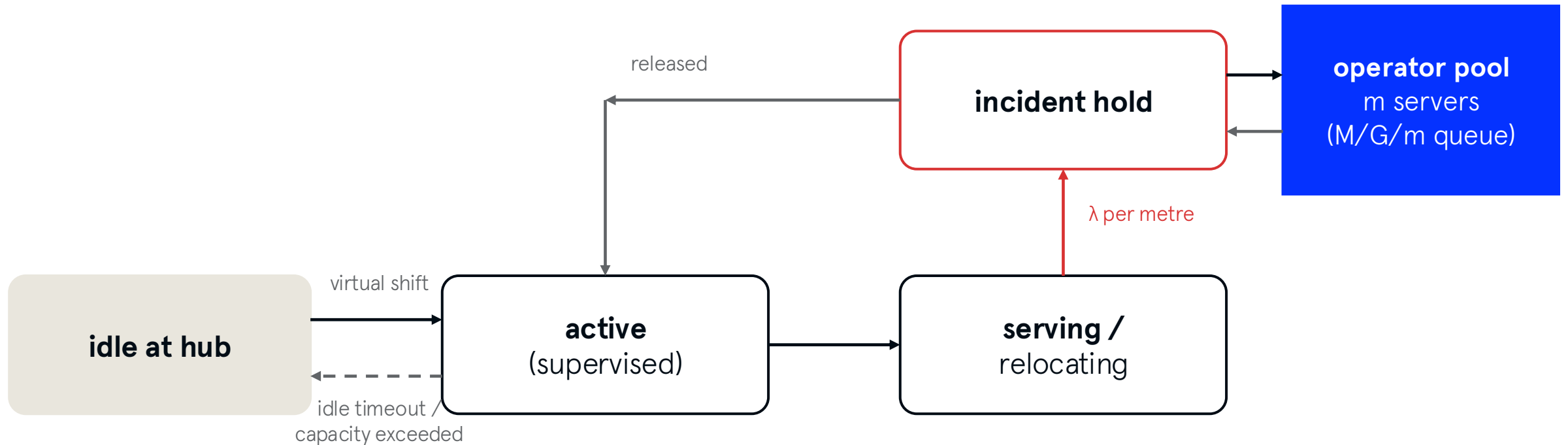
No operational duties
Vehicles always available
Capacity = fleet size

manual

level of automation

autonomous

Three mechanisms, one vehicle life cycle



1 Capacity gating

$$N_{\text{active}}(t) \leq \kappa \cdot L_R(t)$$

Vehicles enter service through transient virtual shifts; pluggable triggers decide how many.

2 Incidents

One Poisson process per severity class over driven metres. The vehicle is held until resolved.

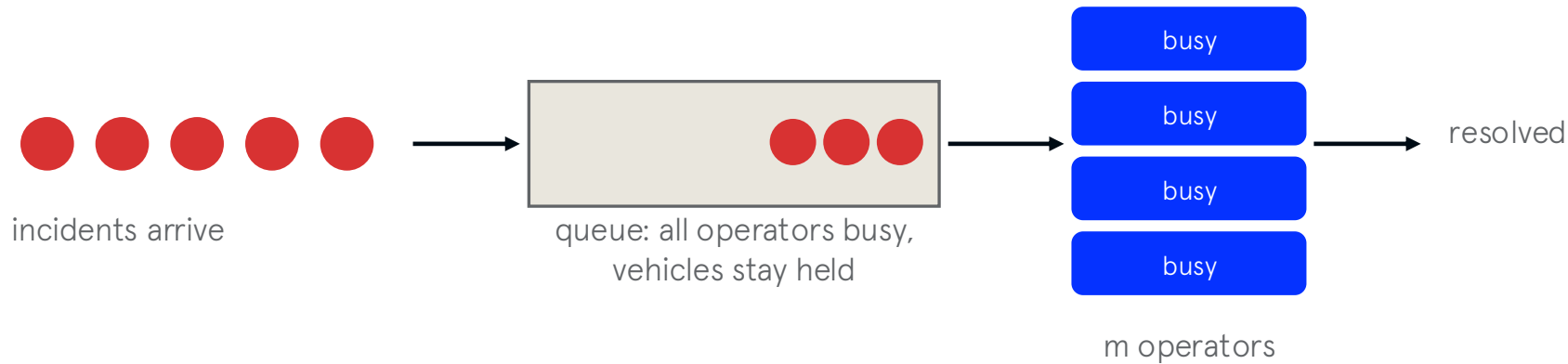
3 Guidance queue

Each incident occupies one operator; when all are busy, requests queue. The pool is an M/G/m system.



Remote Guidance Incident Hold

The operator pool is a queue: three quantities are enough



Λ arrival rate

incidents per hour
 λ per metre $\times$ fleet km per hour

$E[S]$ service time

mean handling time per incident
 45 s in the literature mixture

m servers

operators on shift
 each handles one incident at a time

Offered load $a = \Lambda \cdot E[S]$ (busy operators on average)

Utilisation $\rho = a / m$

Erlang C gives the share of incidents that must wait, in closed form, from a and m alone. At equal ρ a larger pool waits less: the economies of scale of pooling.

Example

Manhattan, 20:00–21:00

$\Lambda = 1,897$ incidents/h
 $E[S] = 45$ s

$a = 1,897 \times 45 / 3,600$
 $= 23.8$ busy operators

$m = 30 \rightarrow \rho = 0.79$

Erlang C: 16 % wait
 simulated: 17 %

The arrival rate is an output, not an input

$$\Lambda(t) = \lambda \cdot dVKT/dt \approx \lambda \cdot N_{\text{active}}(t) \cdot \delta(t) \cdot \bar{v}(t)$$

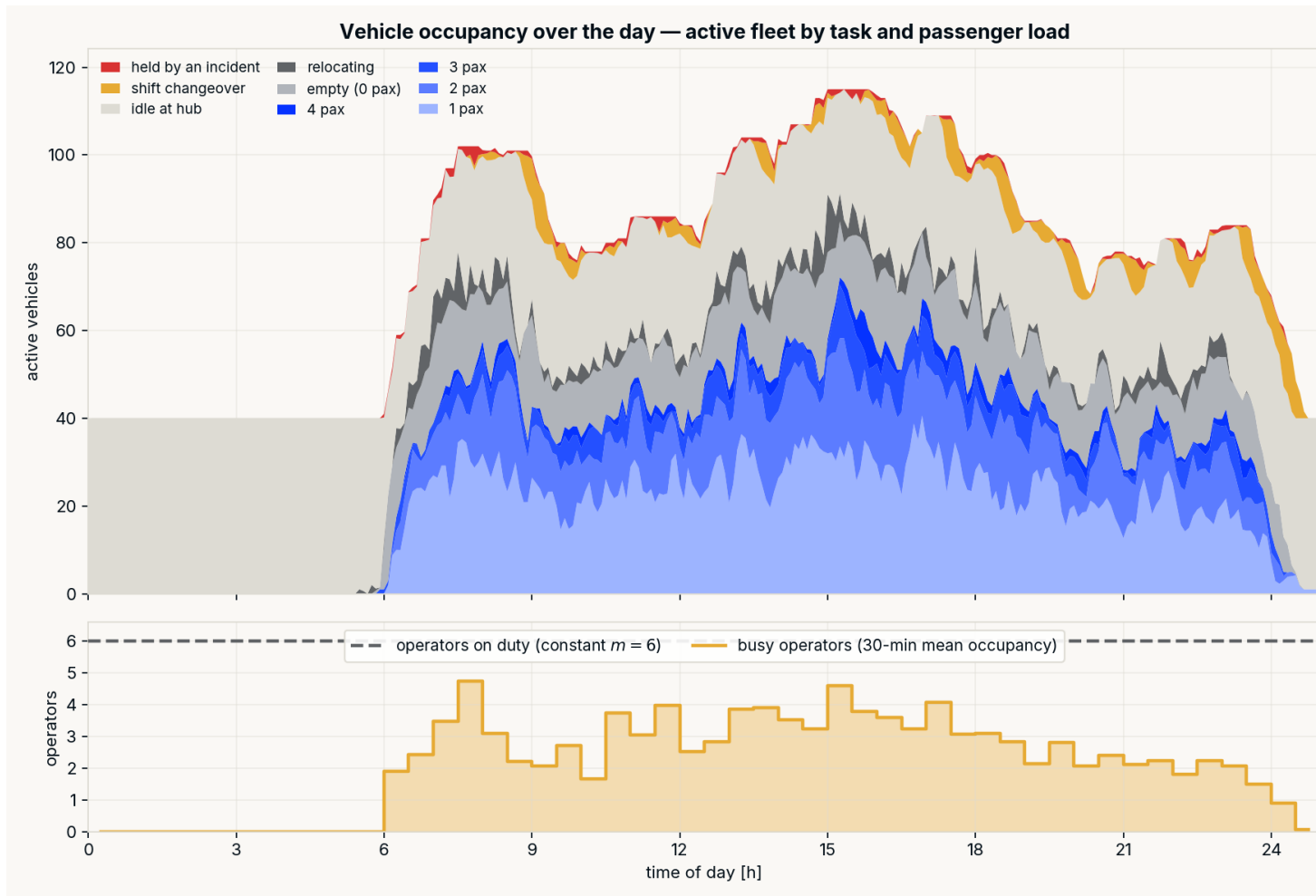
λ is a hazard per metre, a proxy for the maturity of the automated driving system. $\Lambda(t)$ is what the queue sees, in incidents per hour.

$\Lambda(t)$ needs three quantities a closed-form model cannot supply: active vehicles N_{active} , driving share δ and mean speed $\bar{v}$. All three emerge from demand, pooling, routing, empty mileage and the capacity gate.

Held vehicles stop driving, so the arrival rate falls while the queue is long (self-throttling).

Daily incident count = $\lambda \times VKT$, independent of speed. Speed only shifts when incidents happen, and that is what drives contention.

One day in Hamburg: the fleet follows demand, operators gate it



Top: active fleet by state, with incident holds in red.

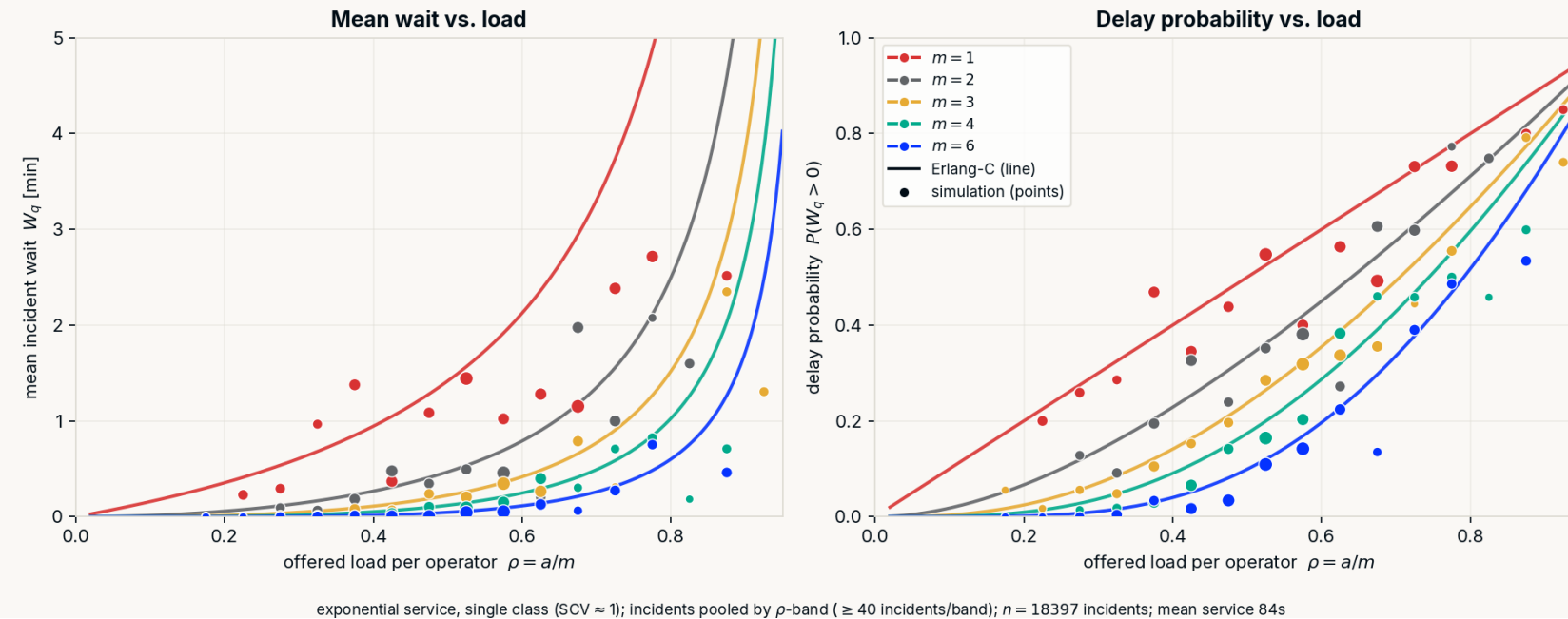
The active fleet grows into the morning, peaks at midday and afternoon, and shrinks at night.

Bottom: the constant pool of six operators, occupancy well below the ceiling.

Incidents act as a time-varying tax on the active fleet, widest when vehicle-kilometres peak.

Verification: the pool reproduces Erlang C where its assumptions hold

Operator queue follows Erlang-C (M/M/m) across the load range and operator ladder



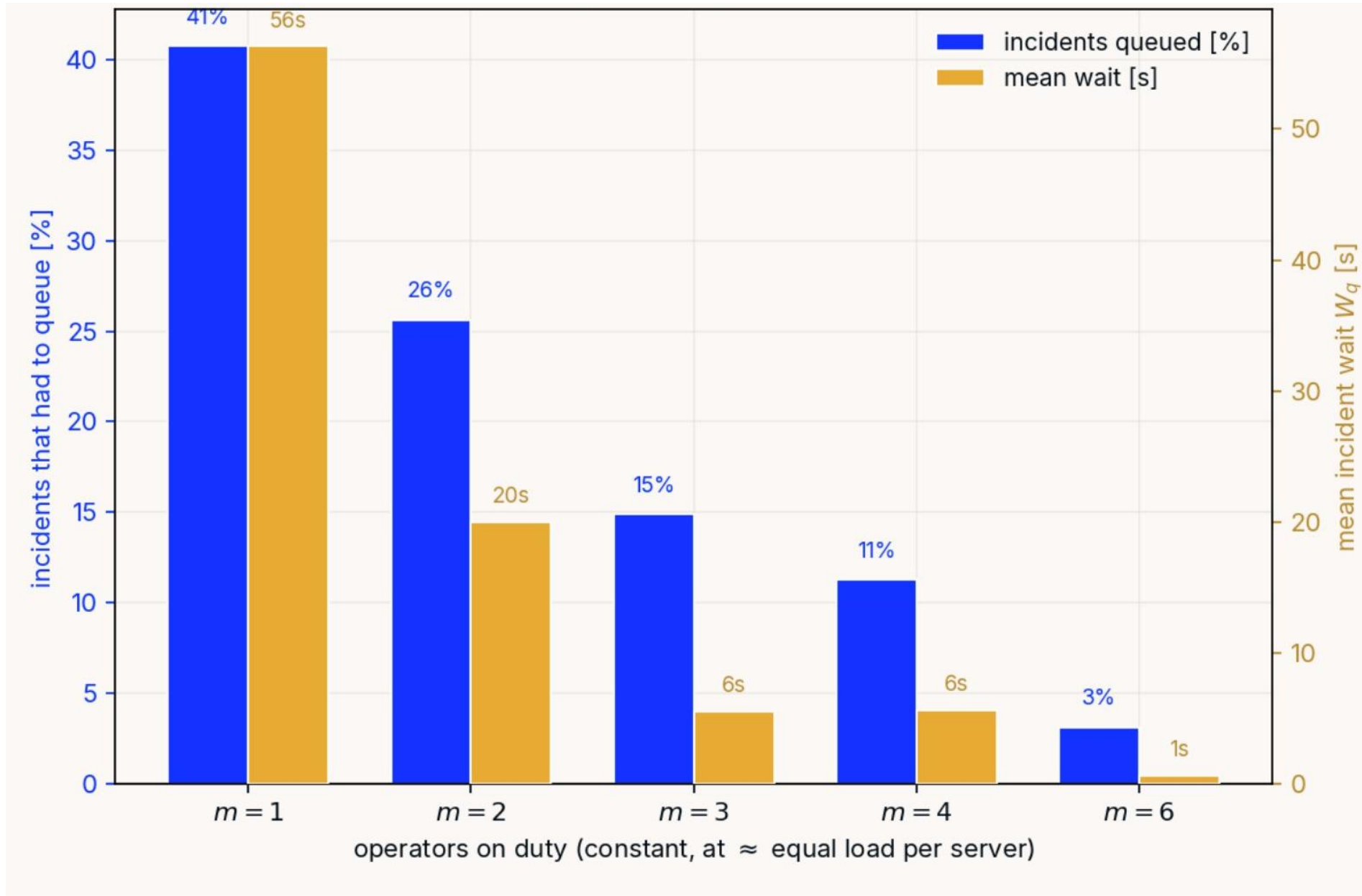
Controlled setting: exponential service, one severity class, fixed m . The pool must then behave as M/M/m.

Ladder $m = 1 \dots 6$, no free parameters.

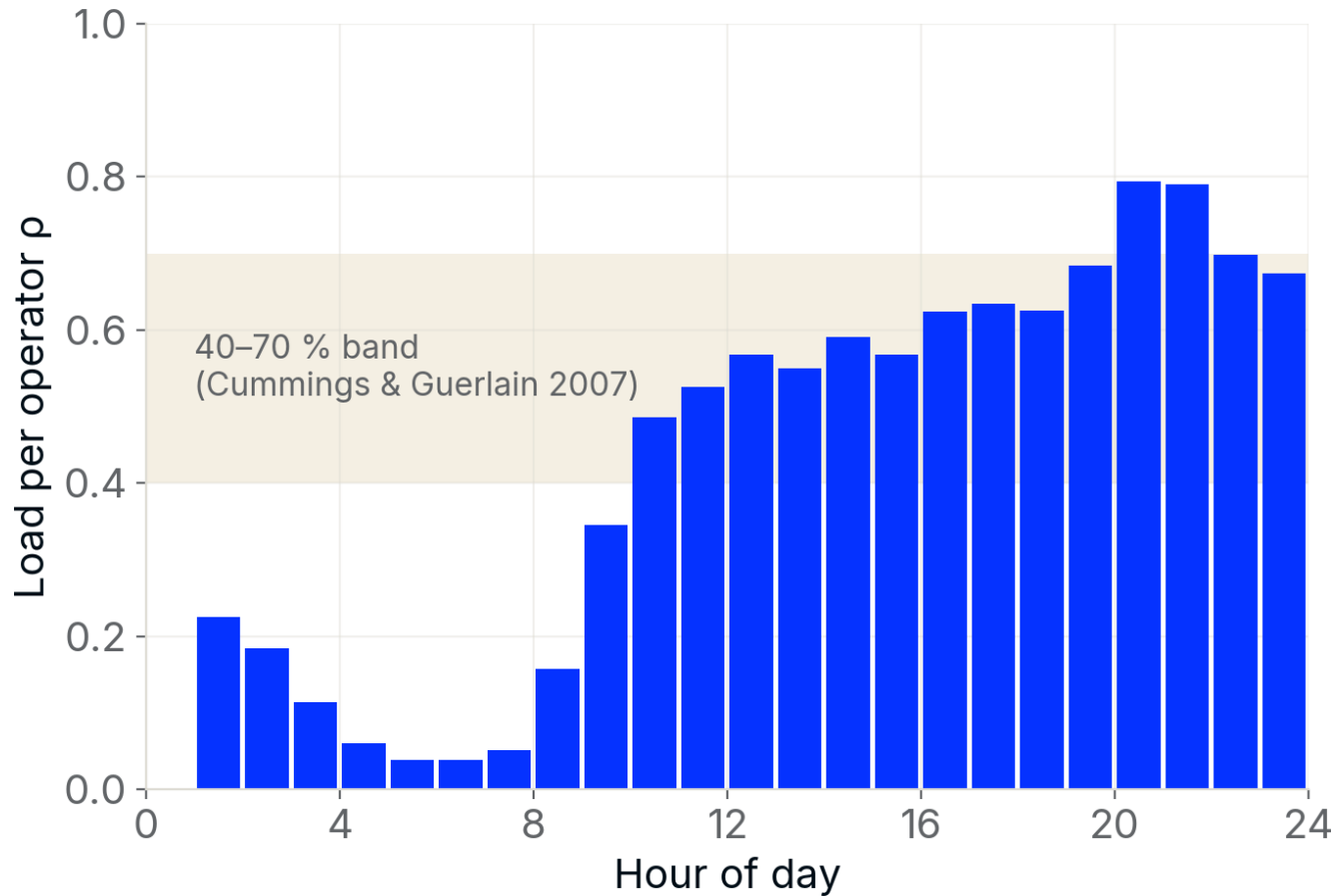
Delay probability tracks Erlang C up to $\rho \approx 0.9$.

Mean wait falls below it: the queue never relaxes to steady state within a day, and the endogenous arrivals are less bursty than Poisson.

Economies of scale: larger pools queue less at equal load



Scaling up: Manhattan with a public scenario



81,319 taxi rides in one day, evening peak, ~170,000 vehicle-km.

Reproducible: scenario from the DRT library, module in the MATSim PR.

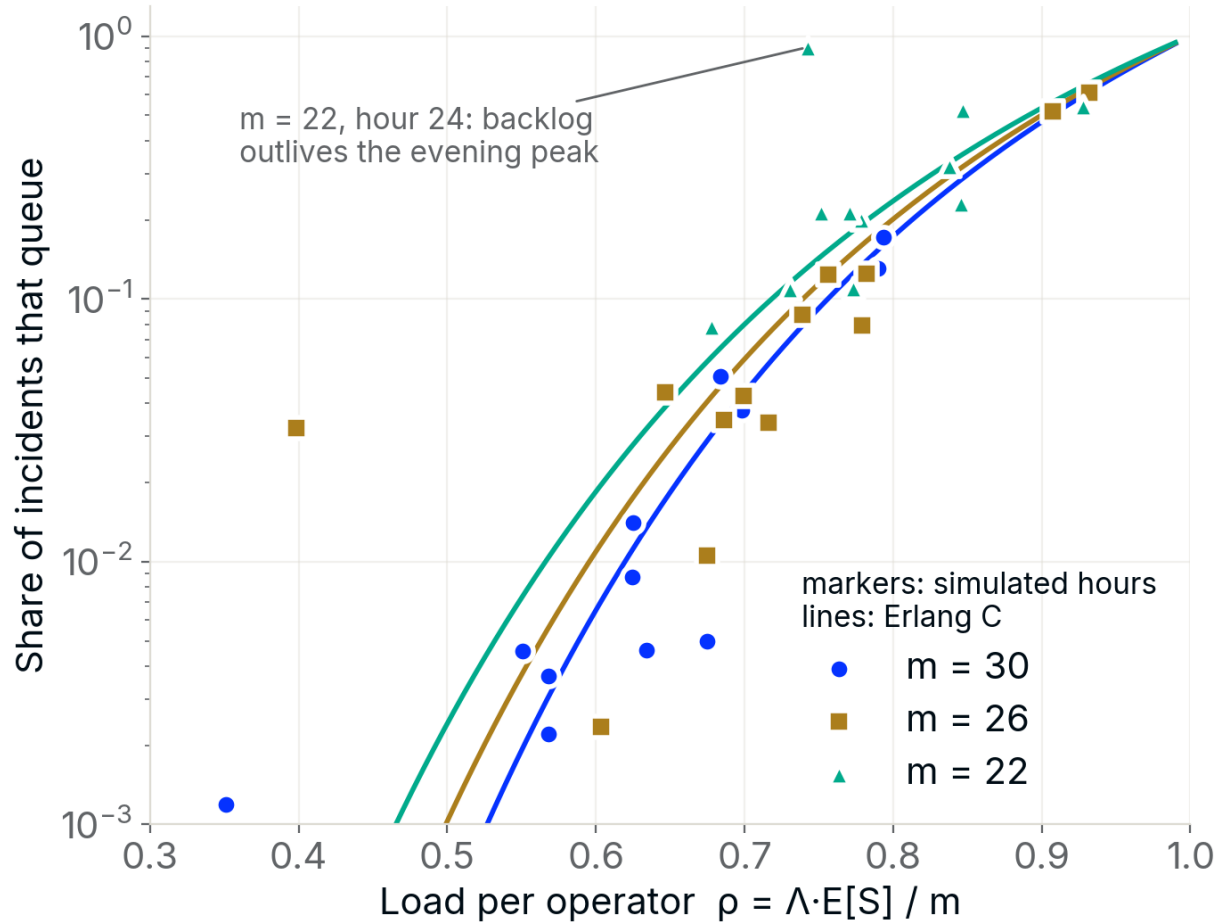
Incident durations from the literature: three classes, 38–58 s, mean 45 s (Schränk et al. 2024; Brecht et al. 2024).

Stress point at the edge of published values: one incident per 7 km, 24,811 incidents.

Load per operator ranges by a factor of 20 within the day at a flat roster.

Cummings, M. L., & Guerlain, S. (2007). Developing Operator Capacity Estimates for Supervisory Control of Autonomous Vehicles. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 49(1), 1–15. <https://doi.org/10.1518/001872007779598109>

Small queues on average, long tails at the peak



Every simulated hour is a point on the Erlang curve, unfitted, at three staffing levels.

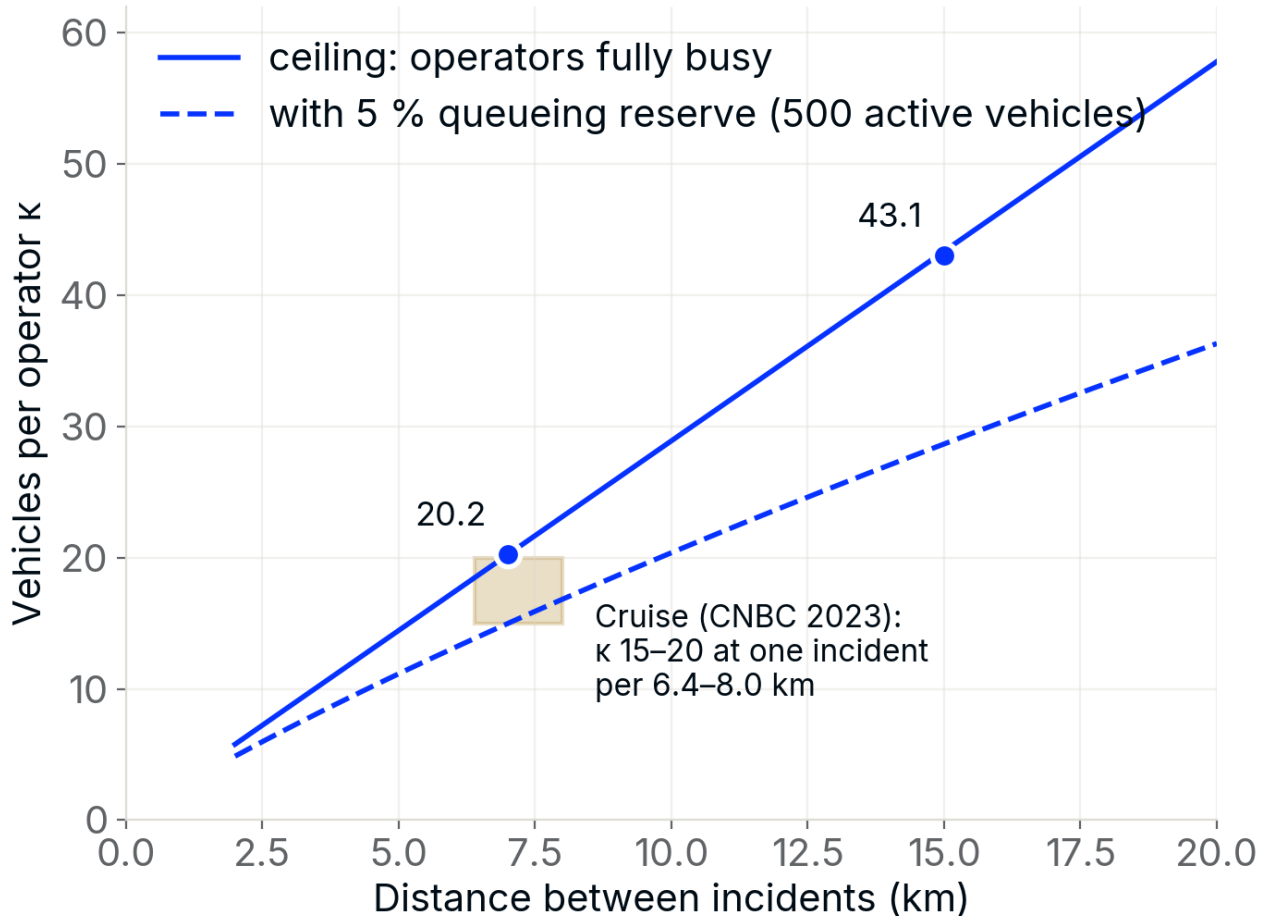
30 operators: 3.2 % of incidents queue (17 % in the peak hour), but the longest wait of the whole day is 28 s.

A mean-wait target does not bind. Short service times and many servers clear a backlog within seconds.

Where the formula fails: hysteresis. At 22 operators the backlog outlives the evening peak; in hour 24 the load is back at $\rho = 0.74$, yet 91 % of incidents queue.

A mean hides the tail. At 26 operators the mean wait is 2 s, yet 22 % of incidents in the peak hour wait longer than 30 s.

κ is a result, not an input



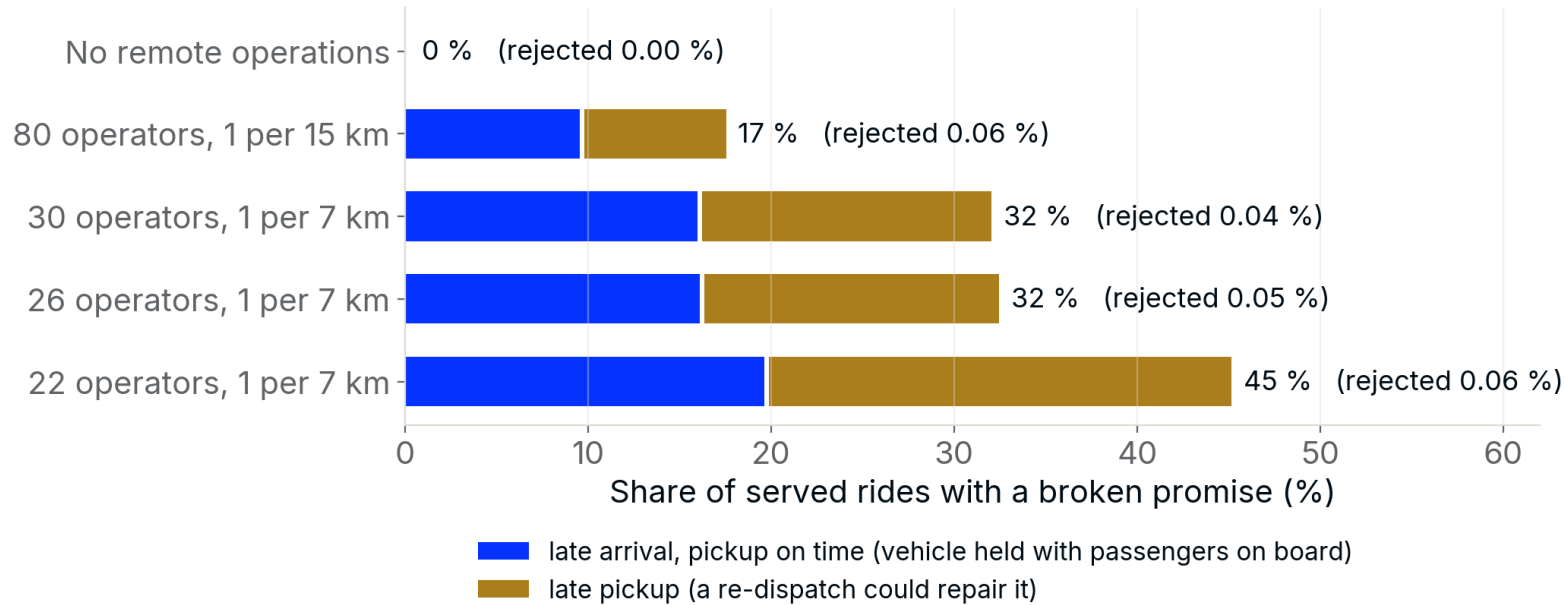
Measured ceiling: 20.2 vehicles per fully busy operator at one incident per 7 km, 43.1 at one per 15 km.

Scales with $1/\lambda$: the 7 km run predicts 43.4 for 15 km.

Geometry behind it: distance between incidents divided by the distance a vehicle drives during one handling.

Resolves the Cruise figures: $\kappa = 20$ at one incident per 6.4–8 km is the ceiling, $\kappa = 15$ is a staffing with 5 % queueing reserve.

Where it reaches the passenger: broken promises, not rejections



Rejections stay below 0.1 %. The infeasibility does not disappear, it moves from admission into execution.

17–45 % of served rides miss the pickup or arrival time promised at booking.

On-board delays respond little to staffing (16.0 % → 19.6 % from 30 to 22 operators). They follow λ and the hold duration.

More operators cut late pickups (25 % → 16 % from 22 to 30), down to a floor set by the hold itself. Below that floor only a lower λ , faster handling or re-dispatch help.

Using the module

```
<parameterset type="drtOperations">
  <parameterset type="shifts"> ... </parameterset>
  <parameterset type="remoteGuidance">
    <param name="defaultOperatorCapacity" value="20"/>
    <param name="activationPolicy" value="buffered"/>
    <param name="readyBufferSize" value="50"/>
  </parameterset>
  <parameterset type="incidents">
    <parameterset type="incidentSeverity">
      <param name="lambdaPerMeter" value="1.4E-4"/>
      <param name="durationMu" value="3.6"/>
      <param name="durationSigma" value="0.5"/>
    </parameterset>
  </parameterset>
</parameterset>
```

Opt-in inside the existing operations extension; operator shifts are ordinary shifts with a dedicated type.

Works with electric fleets and charging, no special cases.

Events for activation, recall, operator shifts and every incident.

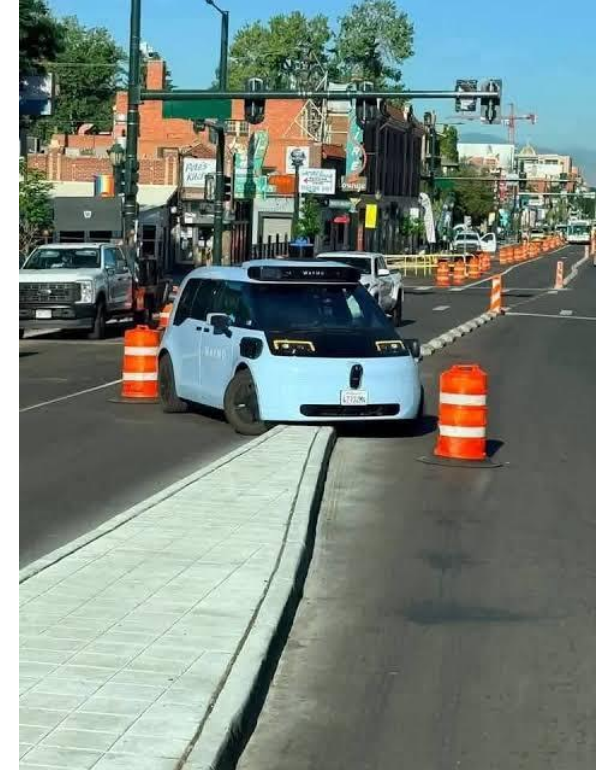
Analysis per iteration: incidents, operator utilisation, active vehicles, operator-hours, incident hotspots (GeoPackage).

matsim-org/matsim-libs #5170 (open): package README and an integration test on the Holzkirchen example.

Outlook and references

- **Dynamic versus static staffing at equal operator-hours**, with the 40–70 % utilisation band as the objective.
- **Ride-hailing versus ride-pooling**: does pooling change the supervision economics?
- **Passenger-side propagation** with a tightened promise and several seeds.
- **In-field assistance** as a second resource for incidents that cannot be solved remotely.
- **Cabin-safety and departure check** at trip starts
- **Incident priorities**

- **Working paper (accepted for TRB)**: Pfeil, Kuehnel & Loder (2026) Smooth Operator: Tracing the Capital-Labor Production Frontier of Remote Operations in Autonomous Mobility-on-Demand. mediatum.ub.tum.de/doc/1859375
- **Shift architecture**: Zwick, Kuehnel & Hörl (2022), Transportation Research Part A 165, 300–320.



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Thank you

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